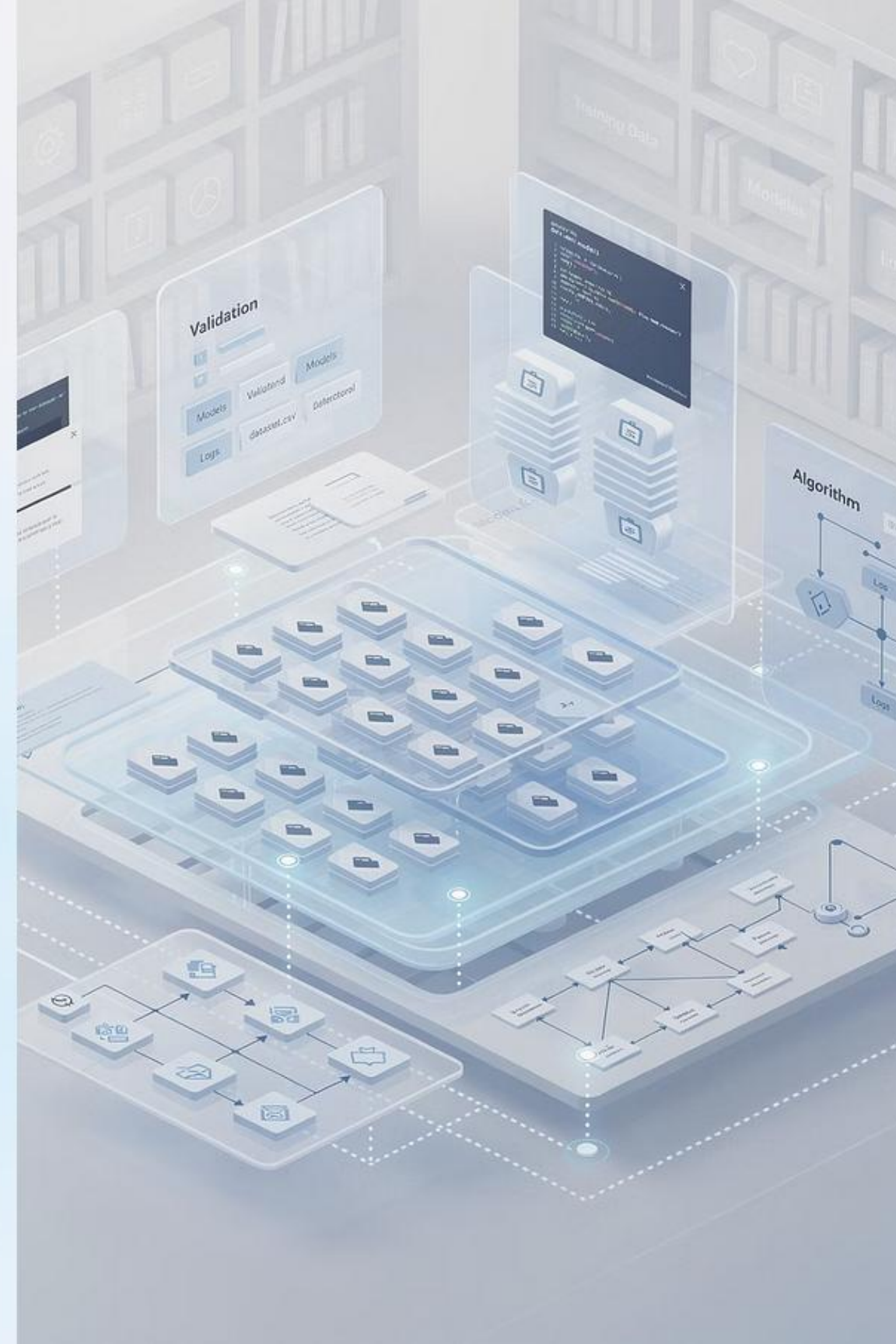


Erasmus Mundus Joint Master Artificial Intelligence for
Sustainable Societies (AISS)

Topics on Machine Learning and its Applications

Repositories in Machine Learning

Pedro Strecht



Repositories in Machine Learning

Main characteristics and best practices

General repositories

Government repositories

Specialized repositories

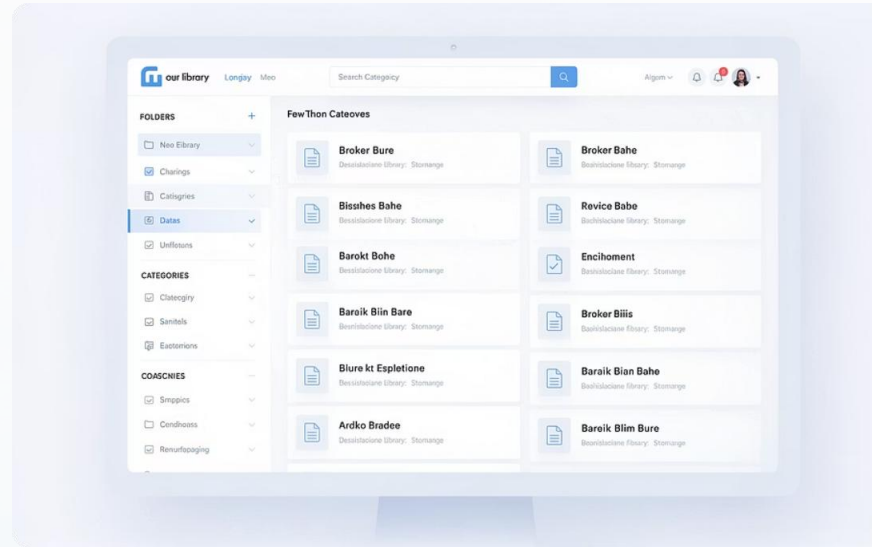
Commons datasets used in research

Main characteristics of data repositories



High-Quality Datasets

Machine learning algorithms depend on large, **high-quality datasets**. These datasets provide the essential experience for algorithms to learn and generalize effectively.



Curated Collections

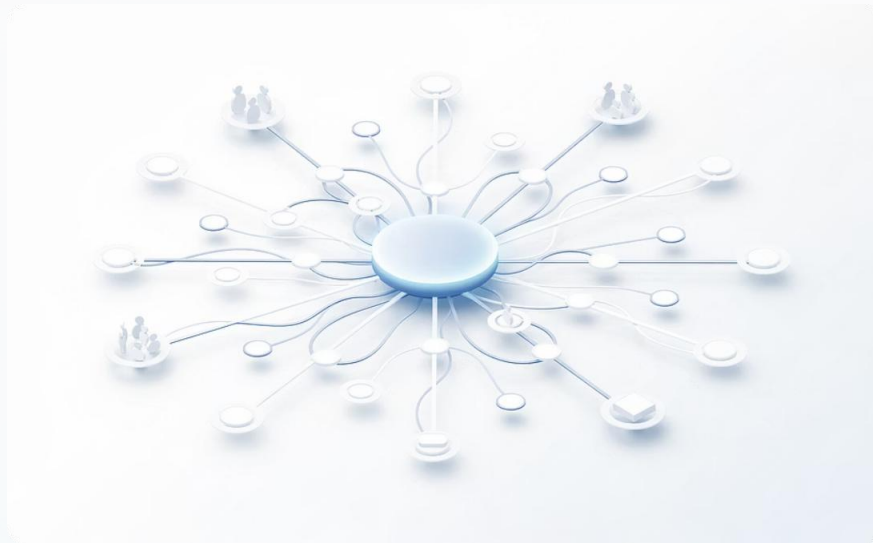
Data repositories offer **curated collections** specifically designed for various machine learning tasks, ensuring relevance and usability.



Diverse Datasets

Access to **diverse datasets** is crucial for fostering robust research and development in the field of machine learning.

Understanding Machine Learning Repositories



Critical Hubs

Serve as critical hubs for the AI community.



Organized Collections

Provide organized collections of datasets, models, and code.



Facilitate Innovation

Facilitate research, development, and reproducibility.

Best Practices for Repository Use



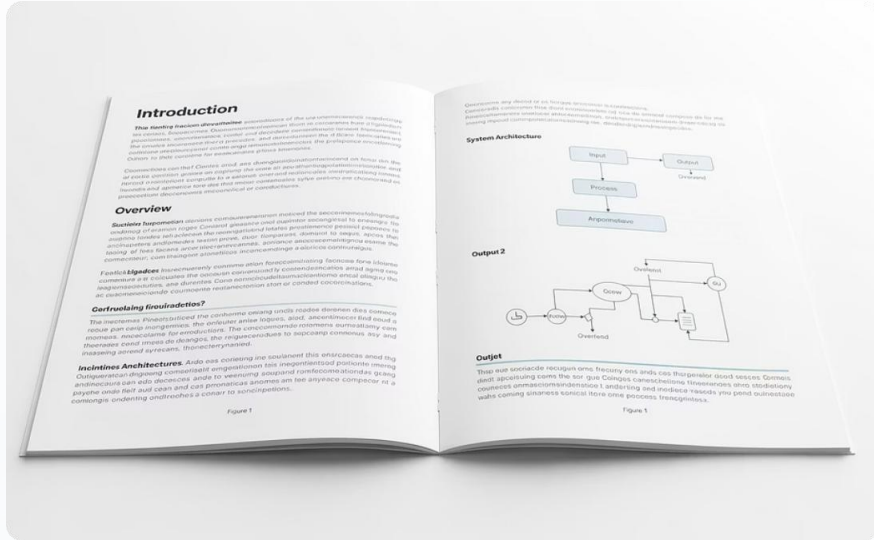
Verify Data Quality & Licensing

Ensure all data used is of high quality and properly licensed for your intended use, preventing legal and ethical issues.



Contribute to the Community

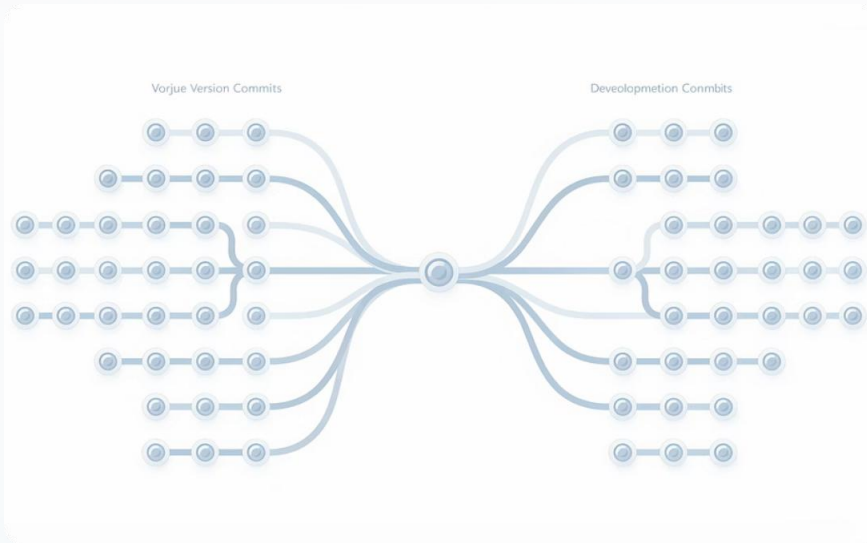
Actively participate by sharing models, datasets, and insights, fostering a vibrant and supportive ecosystem.



Provide Clear Documentation

Thoroughly document your contributions, including methodologies, data sources, and model parameters, for clarity and reproducibility.

Best Practices for Repository Use



Maintain Version Control

Implement robust version control for datasets and code, allowing for tracking changes and collaborative development.



Address Ethical Considerations

Consciously evaluate the ethical implications of your work, especially concerning fairness, bias, and societal impact.



Handle Sensitive Data Responsibly

Apply stringent security and privacy measures when working with sensitive data to protect individuals and organizations.

General repositories

Kaggle



<https://www.kaggle.com/>

Unmatched popularity

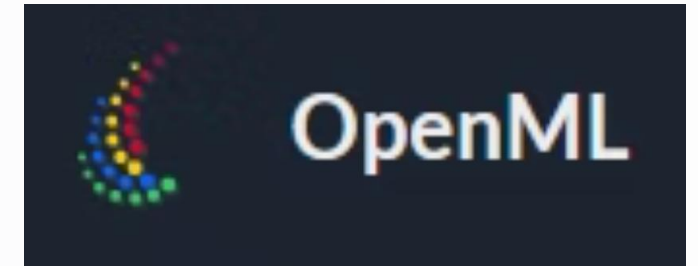
- Kaggle is the world's largest data science community, hosts a vast and ever-growing public repository of datasets
- It is a go-to resource for both aspiring and seasoned data scientists

Fueling competitions

- Kaggle is renowned for its exciting data science competitions
- Many of these competitions leverage datasets from this repository, challenging users to devise innovative solutions for real-world problems

General repositories

OpenML



<https://www.openml.org/>

Meta-learning focus

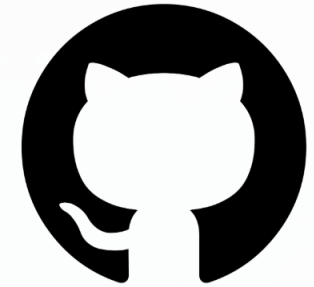
- OpenML facilitates meta-learning research by providing the infrastructure to study the performance of algorithms across a wide variety of datasets and analyze what factors influence their success

API-driven integration

- OpenML offers APIs for Python, R, and Java, enabling seamless integration with popular machine learning tools and workflows. This allows researchers to incorporate OpenML functionality into their existing projects and automate tasks

General repositories

Awesome Public Datasets Collection



<https://github.com/smuthubabu/awesome-public-datasets>

A curated collection

- Organized list of high-quality, topic-centric public datasets
- resource for researchers, data scientists, and anyone interested in exploring data from diverse fields

Community-driven and updated

- The list is actively maintained and updated by the open-source community, ensuring it remains a relevant and comprehensive resource

General repositories

Google Dataset Search

<https://datasetsearch.research.google.com/>

Dedicated data discovery

Google dataset search is a specialized search engine designed specifically for finding datasets across the web

- It crawls and indexes datasets hosted on various platforms, including publisher sites, government portals, and academic repositories, making them easily discoverable



Direct access

Google Dataset Search provides direct links to the datasets, taking you to the source where you can learn more and download the data

General repositories

Open Data on AWS

Open Data
on AWS Data
Exchange

<https://registry.opendata.aws/>

Cloud-Native data sharing

- The AWS Open Data Registry is a curated collection of publicly available datasets that are readily accessible through AWS resources
- This makes it convenient for users already working within the AWS ecosystem

Diverse data with simplified access

- Earth science – Climate data, satellite imagery, geospatial data
- Life sciences – Genomics, proteomics, biomedical data
- Public sector – Census data, economic data, government publications

General repositories

Microsoft Azure Open Datasets



<https://learn.microsoft.com/en-us/azure/open-datasets/dataset-catalog>

Curated for machine learning

Azure Open Datasets offers a collection of curated public datasets specifically designed to enhance the accuracy and efficiency of machine learning workflows on the Azure platform

Diverse datadomains

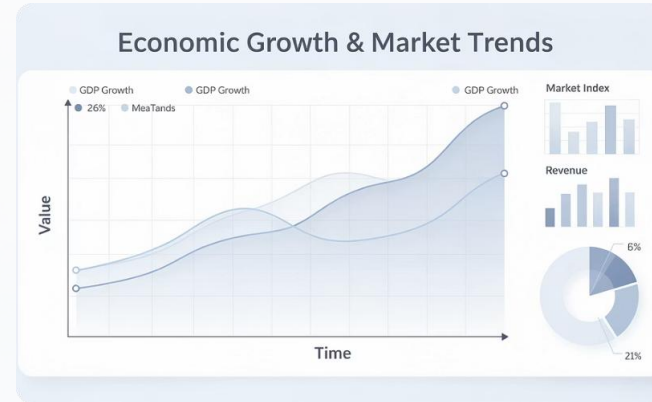
- Weather – Historical weather data for predictive modeling
- Census – Demographic data for socioeconomic analysis
- Holidays – Public holiday data for time series analysis and forecasting
- Finance – Stock prices, economic indicators, and financial time series data

Government repositories



Open Access to Public Data

Government data repositories serve as central hubs for accessing a wide range of data. This data is made available to the public to promote transparency, accountability, and informed decision-making.



Economic Data

Includes employment statistics, inflation rates, and GDP growth for comprehensive financial analysis.



Demographic Data

Covers population censuses, health statistics, and education levels, offering insights into societal trends.



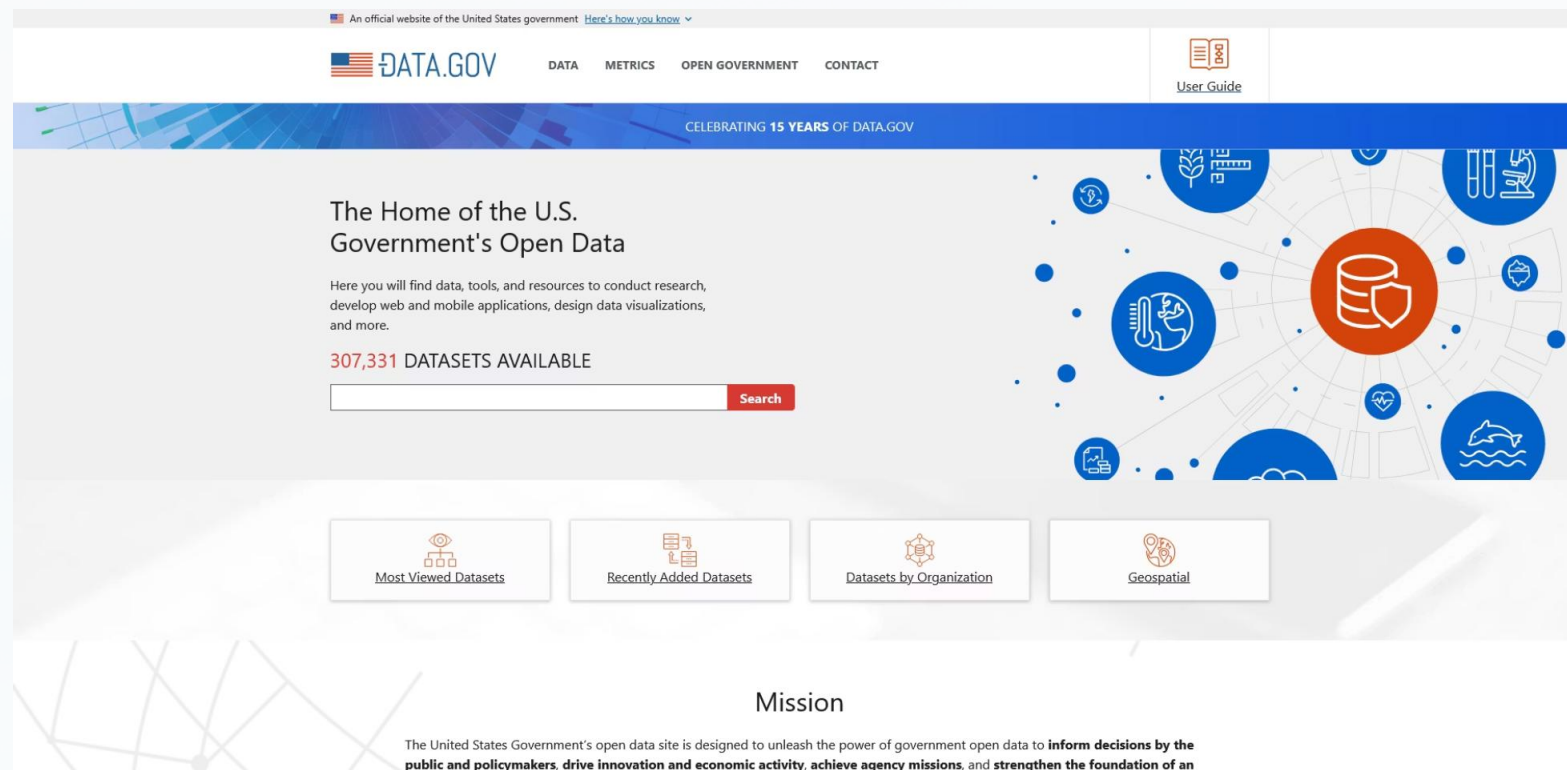
Environmental Data

Features air quality, water quality, and climate data for monitoring ecological health and changes.

Government repositories

<https://data.gov/>

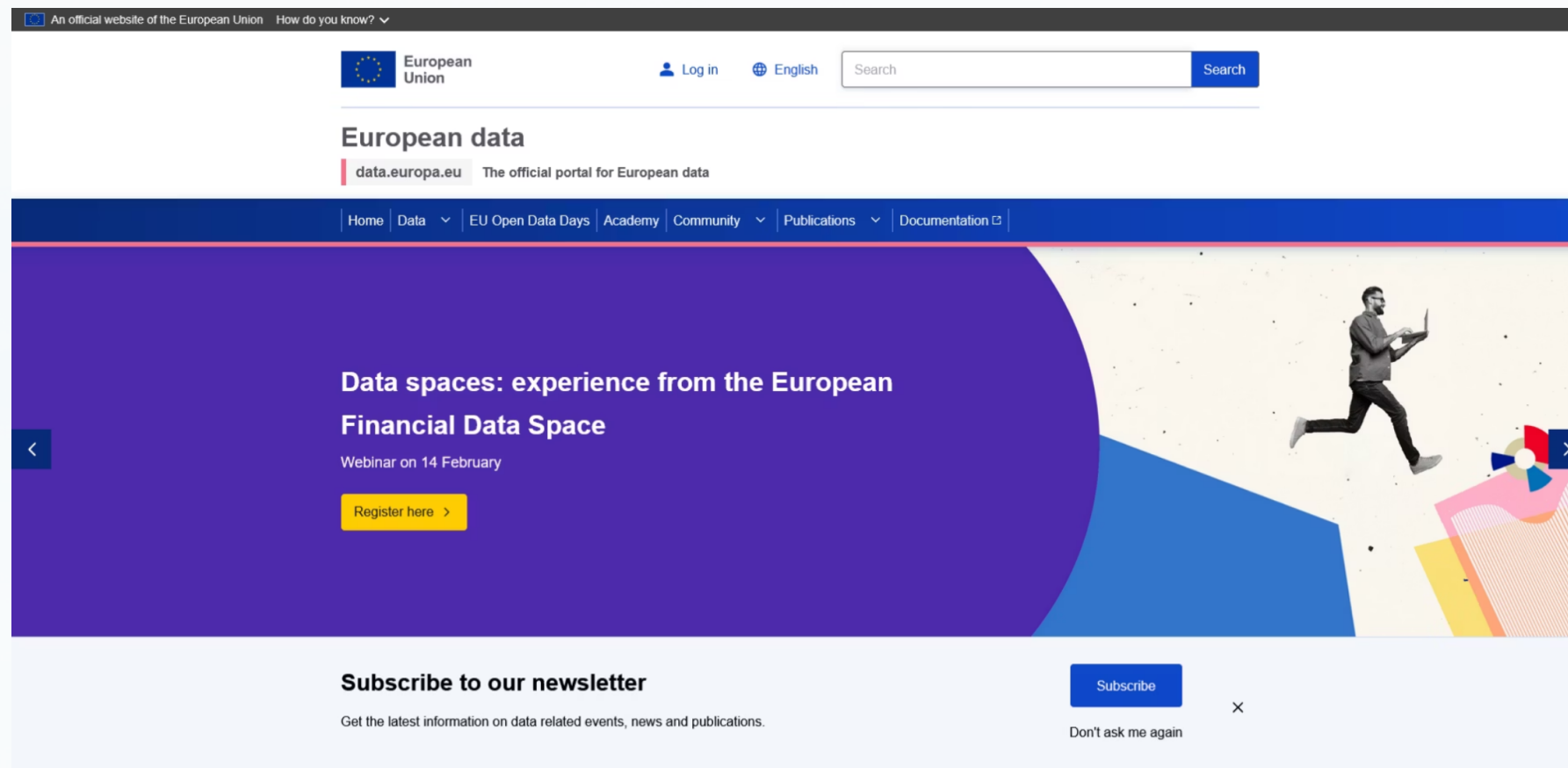
The Home of the U.S. Government's Open Data (USA) – Provides access to data from various US federal agencies



Government repositories

<https://data.europa.eu/en>

The Official Portal for European Data (EU) – Offers access to public data from across the European Union



Government repositories

<https://www.data.gov.uk/>

Find open data (UK) – The UK government's open data portal

data.gov.uk | Find open data

Publish your dataDocumentationSupport

BETA

Find open data

Find data published by central government, local authorities and public bodies to help you build products and services

Search data.gov.uk

[Business and economy](#)

Small businesses, industry, imports, exports and trade

[Crime and justice](#)

Courts, police, prison, offenders, borders and immigration

[Defence](#)

Armed forces, health and safety, search and rescue

[Education](#)

Students, training, qualifications and the National Curriculum

[Environment](#)

[Government](#)

Staff numbers and pay, local councillors and department business plans

[Government spending](#)

Includes all payments by government departments over £25,000

[Health](#)

Includes smoking, drugs, alcohol, medicine performance and hospitals

[Mapping](#)

Addresses, boundaries, land ownership, aerial photographs

[Towns and cities](#)

Includes housing, urban planning, leisure, waste and energy, consumption

[Transport](#)

Airports, roads, freight, electric vehicles, parking, buses and footpaths

[Digital service performance](#)

Cost, usage, completion rate, digital take-up, satisfaction

[Government reference data](#)

Trusted data that is referenced and shared across government

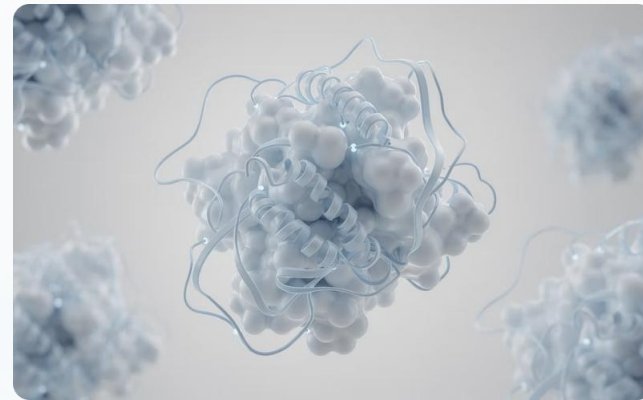
Specialized repositories

Specialized repositories cater to specific disciplines or data types, offering curated collections and tailored services for researchers and practitioners in those fields.



GenBank (Genetics)

A vast collection of genetic sequences



Protein Data Bank (PDB) (Molecular Biology)

A repository for 3D structural data of large biological molecules



ImageNet (Computer Vision)

A large-scale image database designed for visual object recognition research



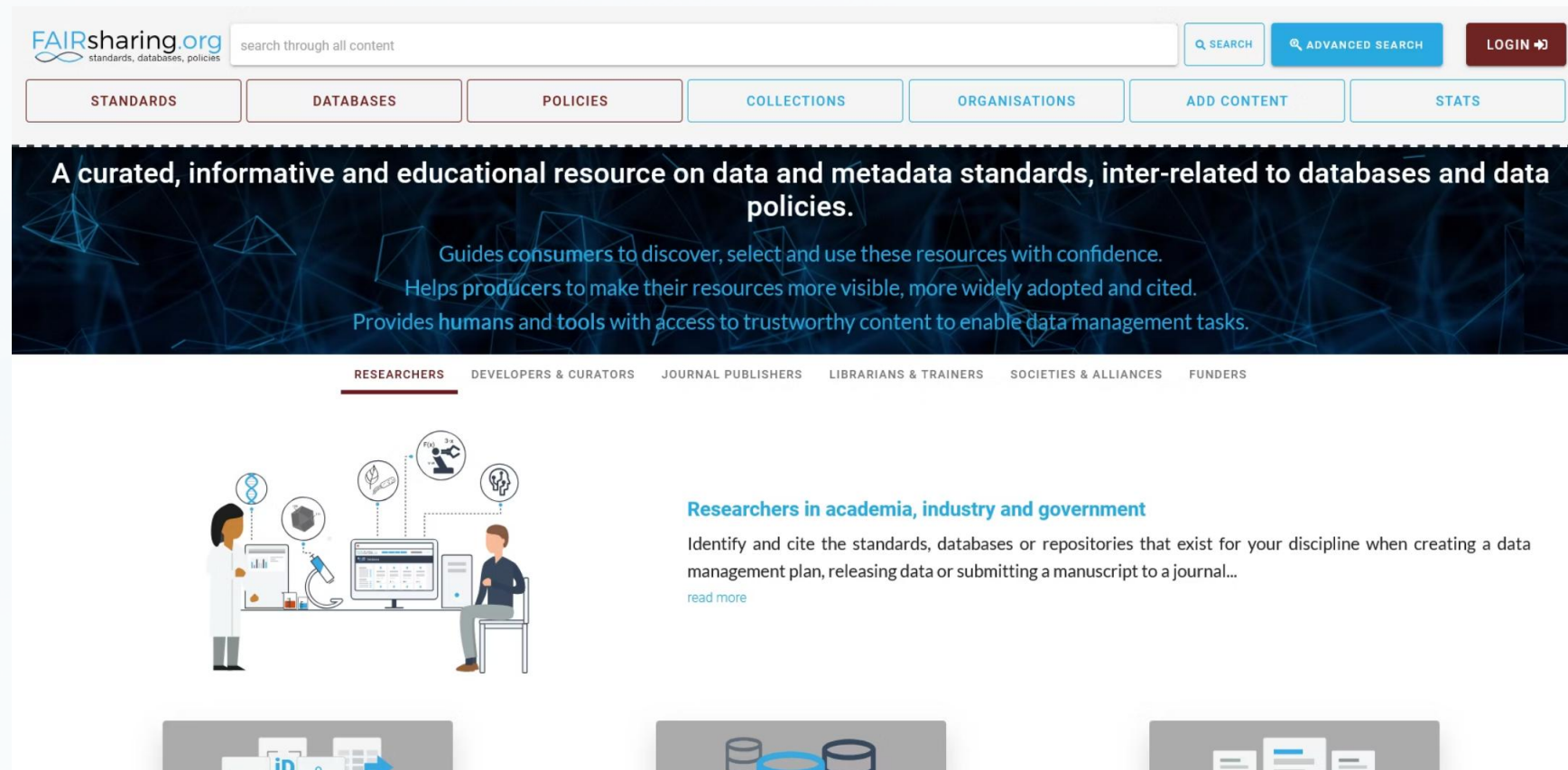
arXiv (Physics, Mathematics, Computer Science)

An open-access archive for electronic preprints of scientific papers

Specialized repositories

<https://fairsharing.org/>

Directory of data assets



Common datasets used in research

For teaching and introductory analysis



Iris

Contains measurements of different iris flower species (used for classification and clustering tasks)



Old Faithful

Records eruption times and waiting times between eruptions of the Old Faithful geyser (useful for regression analysis and demonstrating natural phenomena)



Wine

Chemical analysis of wines derived from three different cultivars (often used for classification and clustering exercises)



Titanic

Contains information about Titanic passengers, including age, class, and survival (widely used for classification tasks)

Common datasets used in research

More complex datasets



Adult (Census Income)

Data from the 1994 US Census, used to predict whether an individual's income exceeds \$50,000 per year based on various attributes. (used for classification, regression, and data exploration)



MNIST

A large database of handwritten digits used for image recognition and classification tasks



CIFAR-10/CIFAR-100

Collections of images commonly used to train machine learning and computer vision algorithms (CIFAR-10 has 10 classes, airplane, automobile, bird, etc. while CIFAR-100 has 100 fine-grained classes)



Boston Housing

Contains information about housing values in the suburbs of Boston (used for regression analysis and predicting house prices)